Observing Physical and Chemical Changes

For each of the following experiments:

1. Give a list of observations for each reaction.
2. Classify each reaction as a physical or a chemical change.
3. Give a reason(s) for why you gave the reaction this classification.

Part A: Dissolve ¼ of a teaspoon of sugar in 50mL of water in a beaker.
Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part B: Burn a piece of magnesium ribbon in a Bunsen burner using tongs. Once it is lit, put it in a crucible. **Safety Note: Do not look directly at the burning magnesium ribbon.**
Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part C: Place an ice block on your hand for between 30 seconds and 1 minute.
Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part D: Place a piece of magnesium ribbon in a test tube then just cover it with Hydrochloric acid. **Safety Note: Acids can burn your skin.**

Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part E: Stir ¼ of a teaspoon of powdered chalk into 50mL of water.
Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part F: Screw up a piece of scrap paper (then place it in the bin)
Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part G: Put 10 drops of Hydrochloric acid into a test tube. Add three drops of phenolphthalein. Slowly add 15 drops of Sodium hydroxide.
Safety Notes: Acids and bases can burn your skin.
Observation:

_____

_____

Physical or Chemical: _____ Reason:

_____

_____

_____

Part H: Put 10 drops of lead nitrate solution n a test tube. Add 10 drops of potassium iodide
Observation:
_____
_____

Physical or Chemical: _____ Reason:
_____
_____
_____

From your observations, list some signs that may indicate a chemical change has occurred.

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